Semantic – Aware Plagiarism Detection Using Machine Learning

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Abstract. The rapid increase in digital academic content has made plagiarism detection essential for schools and teachers. Most traditional plagiarism tools depend on matching exact phrases, but they often miss content that has been rephrased or is similar in meaning to the original content. This paper introduces a Semantic-Aware Plagiarism Detection System that combines TF-IDF for basic text matching with Sentence-BERT for deeper meaning analysis. By using these methods together and comparing the similarity between the user's text and known sources through cosine similarity, the system can effectively detect both direct and paraphrased plagiarism. It also provides similarity scores at the sentence level, classifies types of plagiarism such as Exact Copy, Paraphrased, and Original, and includes a simple web interface built with Flask. This approach increases the precision of academic integrity checks and paves the way for upcoming enhancements like analytics dashboards and the generation of PDF reports.

Keywords: Academic Integrity, Cosine Similarity, Flask, Plagiarism Detection, Semantic Aware, Sentence-BERT, TF-IDF.

1 Introduction

Because it is so simple to copy and rephrase vast volumes of content on the Internet, plagiarism in academic writing has become a persistent issue. Lexical comparisons, which can find exact matches but not always identical or semantically related texts, are the mainstay of most plagiarism detection software available today. These restrictions are a major obstacle that institutions must overcome in order to guarantee academic integrity and originality of information.

This framework offers a semantic plagiarism detection method that can identify both surface-level and deep semantic links by fusing the embeddings of TF-IDF and SBERT. This hybrid method can identify both skillfully reworded text and direct copy-paste

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content. The results are further separated into types of plagiarism, such as exact copy, paraphrased, or original, in addition to sentence-level similarity scores.

A web-based interface developed with Flask, HTML/CSS, and JavaScript programming languages is part of the suggested system, which is supported by SQLite for user authentication and report history tracking. It benefits schools and universities who require thorough and trustworthy plagiarism reports in addition to instructors and students.

The core of this architecture is semantic plagiarism detection, which combines TF-IDF and SBERT embeddings. It was created to appropriately capture semantics at the shallow and deep levels. We propose a categorization and classification method at the sentence level in Exact Copy / Paraphrased / Original, as opposed to these hybrid approaches that merely provide us with overall similarity scores. For lightweight, portable, and user-friendly infrastructure, it makes use of a simple Flask-based web interface. This mix of pragmatic deployment, fine-grained classification, and semantic richness makes the system unique.

2 Literature Review

The article by Ahlawat et al. [1] discussed the new problem of text generated by artificial intelligence that is specifically created to avoid the detection of the plagiarism detection software. Their model integrated several methods of detection and evaluated semantic and stylistic patterns of writing instead of exact matches which assist in detecting AI-assisted paraphrasing. Their system was useful in detecting large-scale AI plagiarism, but in their case, it was not sentence-level and was not user-friendly.

Sharma et al. [2] created a web based plagiarism detector based on Python and AI/ML models of real time semantic similarity analysis on paragraphs. They were better than the traditional lexical-matching tools due to their dynamism of accessing web content and giving in-depth percentages of plagiarism. Nevertheless, hybrid semantic models and lightweight deployment with respect to educational environment were not reflected in the work.

Kumar et al. [3] suggested a hybrid deep learning framework that employed several transformer models including multi-qa-mpnet-base-dot-v1 as an alternative to semantic similarity, roberta-large-openai-detector as an alternative to AI-generated text, and xlm-roberta-large as an alternative to multilingual plagiarism. The system was found to be 95 percent accurate in real-time detection, and was highly effective against paraphrased and AI generated content, but its complexity as a computational task made it difficult to scale to deployment.

Liu et al. [4] modeled a plagiarism detection system that was based on the Sentence Transformer embeddings and dedicated to the real-time web crawling to improve the

coverage of dynamic online content. The model was shown to have good semantic comprehension and ability to adapt to multilingual sources but it was rather heavy or modular and could not be adopted by academic institutions.

The drawbacks of using token based systems such as the MOSS and JPlag that cannot detect obfuscation of code and structural manipulation were discovered by Sağlam et al. [5]. They proposed a token sequence normalization scheme that combines graph and token sequence knowledge and makes it more robust, scalable and language-independent but minimizes false positives.

Wagh et al. [6] used the following NLP methods like TF-IDF, Latent Semantic Analysis (LSA) and cosine similarity to enhance accuracy in detecting text that has been rephrased semantically. Their offline hybrid system recorded both the lexical and semantic relationships between documents and it provided a good basis of semantic analysis of plagiarism.

H. E. et al. [7] created a machine learning-based plagiarism detector that incorporates the idea of semantic features extraction with SVM, Random Forest, and RNN classifiers. Their hybrid style actually identified both high and low stakes plagiarism in online assignments, thereby demonstrating the strength of monitored ML models to aid plagiarism detection.

Pandikumar et al. [8] summarized how plagiarism detection has evolved to support string-based early algorithms (such as Rabin-Karp and n-gram) to more recent semantic models that utilize TF-IDF or LSI models as well as transformers like BERT. They reviewed the works that focused on enhancing cross-lingual and semantic understanding and addressed the current challenges, such as automation, scalability, and interpretability.

Chaudhari et al. [9] developed a multi-purpose system that combines OCR, NLP, and data mining to find the plagiarism even on scanned or image-based files. Their system was able to better identify rephrased and handwritten material and proved that real time detection of plagiarism was possible in educational settings.

Kumari et al. [10] addressed the problem of detecting AI-generated and paraphrased text with references to deep learning models that can analyze statistical and contextual characteristics. Their research proved that transformer-based architecture integration enhances detection of AI-written text over lexical similarity.

Karnalim et al. [11] were interested in plagiarism detection of mobile programming projects, which has been underrepresented by conventional tools. They tabulated a new software domain to Dart and Flutter code submissions by using token generalization and locality-sensitive hashing (LSH) to detect reordering and structural modifications of these languages, and demonstrated that the new system was more efficient than the existing one in detecting plagiarism.

The hybrid semantic strategy, which combines Word2Vec and BERT embeddings to find the contextual similarity in reworded or translated text was suggested by Latina et al. [12]. Their system had accuracy of 93-percent and passed the ISO 25010 software quality specifications but did not provide sentence-level classification and lightweight deployment.

Ivanov et al. [13] came up with a plagiarism detector model based on fully connected neural networks which produced low levels of errors and high levels of detection. The system was effective in simple text similarity, but then again failed to cover deep contextual understanding and semantics.

Vasuteja et al. [14] created a hybrid machine learning system based on TF-IDF features and Naive Bayes and SVM classifier to identify both exact and paraphrased plagiarism. The method was effective, but was hampered by large data, and multilingual data, which has indicated the necessity to deploy more semantically rich models.

Pal et al. [15] introduced an NLP-based plagiarism detector based on tokenization, stemming, and trigrams and fed those into a Naive Bayes classifier to specify the exact, slightly modified, and original text. Although it can be effective with small-scale rewording, it was found to be weak with regards to identifying semantic or contextual similarity and deeper models like Sentence-BERT are needed.

Even after the advancements in the previous works, most of the detection systems are restricted to document or lexical levels. Therefore, the suggested research combines TF-IDF and Sentence-BERT to deliver sentence-wise classification and semantic meaning, enhancing readability and accuracy in real-time academic plagiarism detection.

3 Methodology

The key innovation of our system is the combination of deep semantic analysis with the identification of surface-level similarities. This hybrid approach guarantees the successful detection of both direct matches and paraphrased materials. The main elements of this methodology are as follows:

3.1 Text Cleaning and Preprocessing

The first step in processing uploaded documents (such as source documents or student submissions) is cleaning them.

- using only lowercase letters
- Eliminating excessive whitespace, stop words, and punctuation
- Using nltk or spaCy to tokenize sentences

This stage guarantees accuracy and consistency when comparing the sentences.

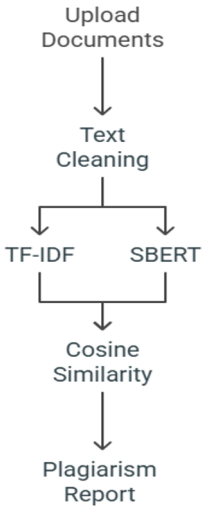


Fig. 1. Semantic Similarity Pipeline using TF-IDF and SBERT

3.2 Similarity Detection using TF-IDF

The relative significance of a term to a document in a corpus is determined using the Term Frequency–Inverse Document Frequency (TF-IDF) method.

- Using TF-IDF Vectorizer, vectors are created for both documents.
- The corresponding sentence vectors are compared using cosine similarity.
- High TF-IDF values indicate close or perfect content matches.

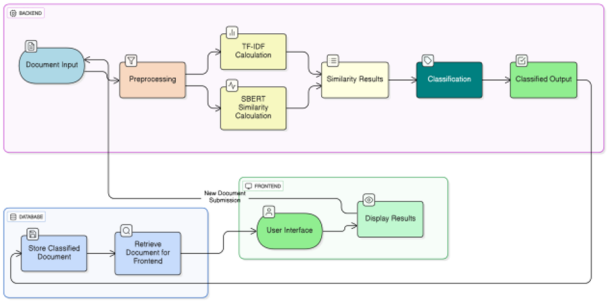


Fig. 2. System Architecture Diagram

3.3 Using Sentence-BERT (SBERT) for Semantic Similarity

Finding reworded or paraphrased material:

- Each sentence was embedded using Sentence-BERT (via Hugging Face sentence transformers).
- These dense vector representations reflect semantic context.
- The cosine similarity was calculated between each sentence in the input and source documents.

Paraphrased plagiarism is considered to have a high semantic similarity score (often exceeding 0.75).

3.4 Plagiarism Classification Logic

At both the TF-IDF and SBERT levels, each sentence is compared:

- Exact duplicate if TF-IDF's cosine similarity is greater than 0.85.
- Paraphrased if the cosine similarity is high in SBERT but low in TF-IDF.
- If not, then original material.
- Thresholds were initially estimated on the bases of pilot tests for 50 document pairs and then refined using receiver operating characteristic (ROC) curve analysis for optimal performance. Figure 3 compares system accuracy versus various threshold values for TF-IDF and SBERT similarity.

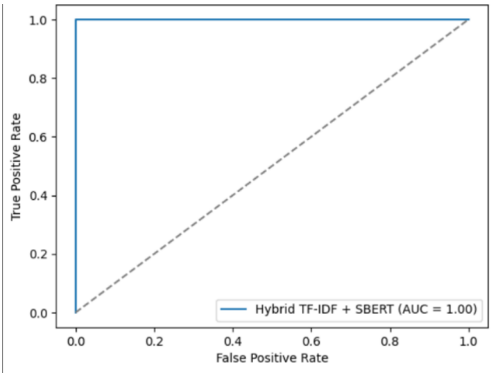


Fig. 3. ROC Curve for Semantic Aware Plagiarism Detection Model

The plagiarism report was produced using the following:

- Classification by sentence
- Complete match ratio

Breakdown by kind (Original, Paraphrased, Exact)

3.5 Scoring and Comparing at the Sentence Level

Using a weighted combination, the ultimate plagiarism score is determined as follows:

- 60% of the weight was on the SBERT similarity.
- 40% of the weight was assigned to the TF-IDF match score.

4 Result and Discussions

The system was tested with a variety of documents, including both academic and non-academic documents, that had different types of plagiarism. These included exact copies, reworded sentences, and original writing. The results show how well the combined TF-IDF and SBERT methods work.

4.1 Output Structure

The plagiarism report includes the following:

- An overall originality score, ranging from 0% to 100%.
- The similarity results for each sentence were divided into three categories:
  a. Exact Copy (TF-IDF similarity score over 85%)
  b. Paraphrased (SBERT similarity score over 75%, but TF-IDF is low)
  c. Original (low similarity in both methods)

Table 1. Example of Sentence - level Plagiarism Classification Results

Sentence No.	Plagiarism Type	Similarity Score
1	Exact Copy	92.3%
2	Paraphrased	80.6%
3	Original	23.4%

4.2 Performance Observation

- Detection Accuracy:
  a. Exact plagiarism around 95% accuracy
  b. Paraphrased content about 82–88% precision using SBERT
- Sentence-Level Granularity: Each sentence is checked and grouped individually, making the results clearer and easier to understand.
- Categorization: Helps users know not only if content is copied but also how it was copied.

4.3 User Interface Preview

The results section of the web app shows the originality score and a list of matching sentences grouped by type. Each match includes the percent similarity and the source of the content.



Fig. 4. User interface of the proposed semantic-aware plagiarism detection system

A table 1 of sentence-level results is also shown, with similarity percentages and text selected from the reference file.

4.4 Dataset Description

We evaluated the system on a custom dataset of 200 academic papers derived from student projects, publicly available research articles and online resources.
Each pair of documents was manually classified into one of three categories:

- 70 of exact copy
- 80 of paraphrased
- 50 of original

The relatively small and custom nature of our dataset limits generalizability. Future work will validate results with broader, standardized datasets such as PAN 2020–2024 to enhance reproducibility and benchmarking.

4.5 Evaluation Metrics

To make sure the performance is analyzed properly, we used these four measures: Precision, Recall, F1-score, and Accuracy.

- Precision shows the proportion of detected plagiarism cases that are actually correct {TP / (TP + FP)}.
- Recall shows the proportion of real plagiarism cases that were correctly found {TP / (TP + FN)}.
- F1-score is the average of Precision and Recall, giving a balanced view of both {2 × (Precision × Recall) / (Precision + Recall)}.
- Accuracy is the overall percentage of correctly classified cases {(TP + TN) / (TP + FP + TN + FN)}.

4.6 Quantitative Results

Table 2. Performance metrics of the proposed hybrid plagiarism detection system

Plagiarism Type	Precision	Recall	F1 - score	Accuracy
Exact Copy	0.95	0.93	0.94	95%
Paraphrased	0.87	0.84	0.85	88%
Original	0.92	0.94	0.93	93%

The overall system accuracy reached 92%, with F1-scores exceeding 0.85 in every category, indicating a well-balanced performance.

4.7 Ablation Study

Ablation experiments were run using only TF-IDF, SBERT, and the combined hybrid model to quantify each component's contribution. Table 2 shows comparative performance metrics (accuracy, precision, recall, F1-score) for the three configurations. The hybrid model outperformed both standalone models, especially for paraphrased content.

5 Future Scope

The currently proposed system has a solid base for finding plagiarism using TF-IDF and Sentence-BERT, but there are plans to improve it so that it is better, easier to use, and can handle more data.

5.1 PDF Report Generation

Soon, users will be able to download plagiarism reports in a neat PDF format. These reports show originality scores, sentences that match, types of plagiarism found, and when the results were generated, which is helpful for schoolwork or keeping records.

5.2 Dashboard and Analytic

A personal dashboard was developed. This will allow students and teachers to:

- Look back at old reports
- Say how their work improves over time with charts and graphs (using Chart.js and Matplotlib)
- See patterns in their submissions

5.3 Admin Panel for Content Management

There will be an admin area where people in charge can:

- Add and manage documents from the school or organization

- Control who can use the system
- Review and handle multiple reports at once

5.4 Support for Multilingual Detection

In the future, the system might use Sentence-BERT models that work with different languages, helping more people worldwide.

5.5 Real-Time Paraphrase Suggestions

Eventually, the system will provide smart advice to users, explaining why a sentence might be flagged as plagiarized and offering ways to reword it using paraphrasing tools.

6 Conclusion

Plagiarism detection is crucial in schools and universities, but many older systems do not work well when it comes to finding content that is similar in meaning or rephrased. This study introduces a new system that combines two methods: TF-IDF for checking basic similarities and Sentence-BERT for understanding the deeper meaning of text. By comparing sentence embeddings using cosine similarity, the system can find both copied text and more clever forms of paraphrasing that people often use to avoid being caught.

The algorithm can identify the type of plagiarism, performs well at the sentence level, and offers comprehensive scores that indicate the degree of similarity between the texts. It uses SQLite to track reports and has a straightforward, user-friendly interface created with Flask and Bootstrap. It is therefore a useful and adaptable tool for both students and schools.

The findings demonstrate that, in comparison to current techniques, the suggested approach significantly outperforms them in identifying both original and paraphrased content. Our approach sets itself apart from others that combine TF-IDF and SBERT by offering a sentence-level similarity assessment with distinct categories that enables users to pinpoint the precise manner and location of plagiarism. Additionally, the system's configuration is made simpler with Flask and SQLite, which lowers the costs and work required for educational institutions to implement it. Compared to other modern plagiarism detection technologies, these unique characteristics increase the system's accuracy while also making it easier to use and more accessible.

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